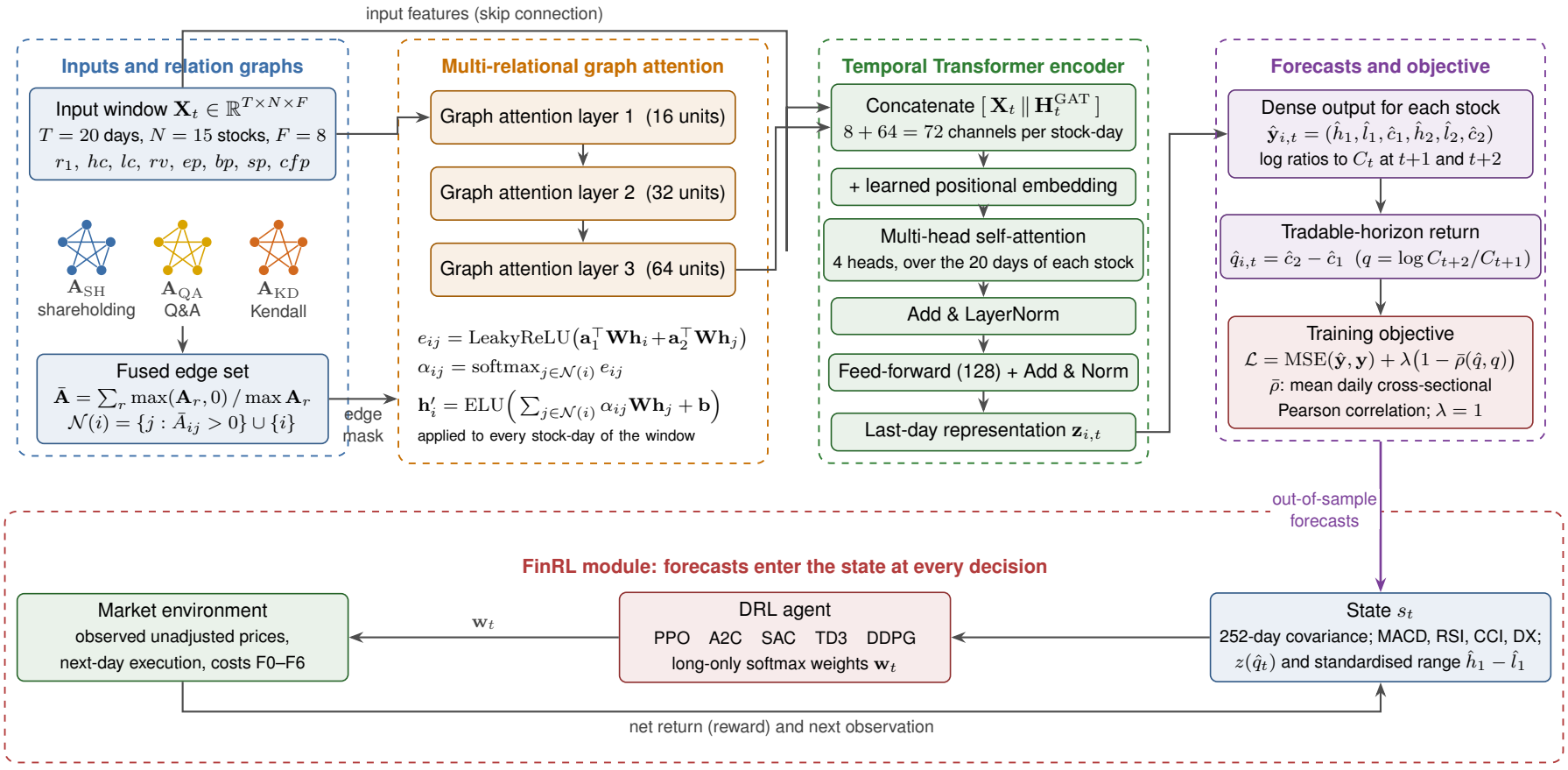

input features (skip connection)
Inputs and relation graphs
Input window $\mathbf{X}_t \in \mathbb{R}^{T\times N\times F}$
$T = 20$ days, $N = 15$ stocks, $F = 8$
$r_1,\ hc,\ lc,\ rv,\ ep,\ bp,\ sp,\ cfp$
$\mathbf{A}_{SH}$ shareholding
$\mathbf{A}_{QA}$ Q&A
$\mathbf{A}_{KD}$ Kendall
Fused edge set
$\bar{\mathbf{A}} = \sum_r \max(\mathbf{A}_r, 0) \,/\, \max \mathbf{A}_r$
$\mathcal{N}(i) = \{j : \bar{A}_{ij} > 0\} \cup \{i\}$
edge mask
Multi-relational graph attention
Graph attention layer 1 (16 units)
Graph attention layer 2 (32 units)
Graph attention layer 3 (64 units)
$e_{ij} = \mathrm{LeakyReLU}\big(\mathbf{a}_1^\top \mathbf{W}\mathbf{h}_i + \mathbf{a}_2^\top \mathbf{W}\mathbf{h}_j\big)$
$\alpha_{ij} = \mathrm{softmax}_{j\in\mathcal{N}(i)}\, e_{ij}$
$\mathbf{h}'_i = \mathrm{ELU}\Big(\sum_{j\in\mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j + \mathbf{b}\Big)$
applied to every stock-day of the window
Temporal Transformer encoder
Concatenate $[\,\mathbf{X}_t \,\|\, \mathbf{H}_t^{\mathrm{GAT}}\,]$
$8 + 64 = 72$ channels per stock-day
+ learned positional embedding
Multi-head self-attention
4 heads, over the 20 days of each stock
Add & LayerNorm
Feed-forward (128) + Add & Norm
Last-day representation $\mathbf{z}_{i,t}$
Forecasts and objective
Dense output for each stock
$\hat{\mathbf{y}}_{i,t} = (\hat{h}_1, \hat{l}_1, \hat{c}_1, \hat{h}_2, \hat{l}_2, \hat{c}_2)$
log ratios to C_t at $t{+}1$ and $t{+}2$
Tradable-horizon return
$\hat{q}_{i,t} = \hat{c}_2 - \hat{c}_1 \quad (q = \log C_{t+2}/C_{t+1})$
Training objective
$\mathcal{L} = \mathrm{MSE}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda\big(1 - \bar{\rho}(\hat{q}, q)\big)$
$\bar{\rho}$: mean daily cross-sectional Pearson correlation; $\lambda = 1$
out-of-sample forecasts
FinRL module: forecasts enter the state at every decision
Market environment
observed unadjusted prices, next-day execution, costs F0–F6
$\mathbf{w}_t$
DRL agent
PPO A2C SAC TD3 DDPG
long-only softmax weights $\mathbf{w}_t$
State s_t
252-day covariance; MACD, RSI, CCI, DX; $z(\hat{q}_t)$ and standardised range $\hat{h}_1 - \hat{l}_1$
net return (reward) and next observation